

A Monte Carlo method to solve the optimal coordination of directional overcurrent protections considering all the possible fault locations

Paulo M. De Oliveira-De Jesus^{a,*}, Elmer Sorrentino^{b,c}

^a Electrical and Electronic Engineering Department, School of Engineering, Universidad de Los Andes, Bogotá, Colombia

^b Department of Electrical Engineering, Universidad Carlos III de Madrid, Calle Butarque 15, 28911 Leganés, Spain

^c Universidad Simón Bolívar, Valle de Sartenejas, Caracas, Venezuela

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ABSTRACT

This paper presents a Monte Carlo method to solve the Optimal Coordination of Directional Over-Current Protections (OC-DOCP) considering transient configurations and all the possible Fault Locations (FLs). Previous methods cannot guarantee complete selectivity because they do not consider all the possible FLs. In the proposed method, near-end, and far-end faults are complemented by additional FLs, which are chosen from many randomly generated candidates (Monte Carlo method). Each FL candidate is included in the set of constraints at once to solve an OC-DOCP, whose results are the inputs for a simulator that assesses speed and selectivity indexes considering all the possible FLs. Thus, the FL with the maximum selectivity index is chosen to remain in the set of constraints for the next steps in an iterative procedure that finishes when the specified level of selectivity is reached. From the first iteration, the selectivity index obtained with the proposed method is substantially better than those obtained with the optimal solution of previous methods. The selectivity index is increased successively until it reaches practically 100%, without the need of including all the FLs in the set of constraints. Furthermore, an important improvement in the selectivity index is obtained without a significant worsening of the protection speed. The fact of solving many small optimization problems is an attractive feature of this method because facilitates the possibility of applying parallel computing to this problem.

1. Introduction

The function of the protection system is to isolate the power system faults as quickly as possible, in order to prevent equipment damage and reduce the risk of power system collapse. Speed is an essential attribute, but sensitivity and selectivity are also fundamental [1]. The protection system should be able to detect the possible faults, irrespective of fault magnitude or location, and provide the adequate tripping to avoid unnecessary outages.

The procedure for coordination of Directional Over-Current Protections (DOCP) to assure adequate speed and selectivity in networks with multiple sources is a challenging task [2]. The use of optimization to cope with the coordination problem was originally proposed many years ago [3] and it can be considered a well-known topic.

Many contributions on Optimal Coordination of DOCP (OC-DOCP) have been proposed in the literature, and a selection of ten of the most cited articles up to now (according to Web of Science) is herein included [3–12]. Furthermore, summarized descriptions of papers about this topic have been published (e.g., [13–17]). A wide diversity of

optimization tools has been applied in the research about this problem, ranging from linear programming methods (e.g., [18,19]) to non-linear programming approaches (e.g., [20]), as well as meta-heuristic techniques, such as evolutionary-based methods (e.g., [21]), genetic algorithms (e.g., [22,23]), differential-search algorithms (e.g., [24,25]), and particle swarm optimization (e.g., [26]). Most of the existing contributions on OC-DOCP are based upon a single objective, that usually is the minimization of a weighted sum of operation times, subject to sensitivity and selectivity constraints (e.g., [27–29]). Although there are also some contributions based on the multi-objective perspective to solve the OC-DOCP (e.g., [30–33]), these approaches are actually looking for the maximization of the protection speed.

On the other hand, some papers indicate that the Transient Configurations due to the Sequential Tripping At Both Line-Ends (TC-STABLE) should be considered in the OC-DOCP formulation (e.g., [34]), but unfortunately this effect has been rarely included. Furthermore, some papers indicate the need of considering different Fault Locations (FLs) in the problem formulation (e.g., [35]) to guarantee selectivity for any

* Corresponding author.

E-mail address: pm.deoliveiradejes@uniandes.edu.co (P.M. De Oliveira-De Jesus).

Nomenclature

Subscripts

c	Close-in fault location
i	Selected fault location
j	Candidate coordination solution
k	Iteration counter
r	Generic DOCP
p	Backup DOCP
q	Primary DOCP
u	DOCP-pair

Superscripts

$+$	Variable for the simulator (not for the linear programming problem)
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Vectors and Matrices

$\mathbf{B}$	Coordination matrix
$\mathbf{D}$	Vector of time dial settings
$\mathbf{D}_{\min}$	Vector of $\mathbf{D}$ lower bound
$\mathbf{D}_{\max}$	Vector of $\mathbf{D}$ upper bound
$\mathbf{H}$	Vector of fault locations
$\mathbf{S}$	Vector of separation times between backup and primary DOCPs

Variables

α_{r1}	Curve parameter of DOCP r
α_{r2}	Curve parameter of DOCP r
β	Term of $\mathbf{B}$ for the line-end that trips first
γ	Term of $\mathbf{B}$ for the second line-end that trips
$D_{\min}$	D_r lower bound
$D_{\max}$	D_r upper bound
D_r	Time dial setting of DOCP r
ϵ_{sel}	Desired level of selectivity
H_i	Fault location i
I_r	Fault current magnitude through DOCP r
n_c	Number of solution candidates
N_F	Number of DOCP pairs where is feasible to compute a separation time
n_{fl}	Number of fault locations
n_{fls}	Number of simulated faults locations
N_{nosel}	Analyzed faults with no-selective DOCP pairs
n_{pr}	Number of total primary-backup DOCP pairs
n_r	Number of DOCPs
N_{sfl}	Number of selected fault locations
P_r	Pick-up current of DOCP r
S_u	Separation time of DOCP-pair u
S_M	Safety margin
T	Operation time

FL, but unfortunately the OC-DOCP have been frequently formulated in many papers with only one or two fault locations (only near-end, or only near-end and far-end).

There are different published OC-DOCP solutions for a given sample system, mainly because the premises for the optimization model are different from each other. It was shown that all of them can fail in guaranteeing the selectivity for all the FL [36]. The main reason behind this fact is that the different FLs were not included in the set of constraints. In [36], a simulation tool was developed in order to assess indexes

of selectivity, speed and sensitivity of any coordination solution, by computing the performance of the DOCP for 1000 uniformly distributed FLs along each transmission line, considering TC-STABLE. For instance, a published solution [37] for a well-known example [38] does not have good selectivity because does not consider all the possible FLs nor TC-STABLE [36]. The published solutions can also fail in guaranteeing sensitivity for solid faults in all the possible FLs [36], but this problem is not herein analyzed because it is assumed, as in the cited example [37], that the sensitivity of all the DOCP has been previously maximized, avoiding wrong tripping due to normal load conditions.

The coordination of protections intrinsically is a multi-objective problem [39]. That is, an increase of the afore-mentioned selectivity index, considering all the FL, should surely imply a decrease of the protection speed for the solution of the OC-DOCP [39]. On the other hand: (a) the simultaneous inclusion of all the FLs in the set of constraints would increase enormously the mathematical problem dimension; (b) although there are methods to deal with TC-STABLE based on linear programming, they require an iterative procedure since the sequence of trips at both line ends cannot be known a priori [18,28,34]. Thus, the OC-DOCP considering all the FLs and the TC-STABLE is a challenging problem. This problem had not been previously solved in the literature.

To solve this problem, we propose an optimization method that includes only a limited number of FLs (in per unit of line length), which are selected using Monte Carlo simulation in order to find the best FL to be included in each step, from a set of randomly generated candidates. An OC-DOCP is solved, including each candidate at once, and the best FL is the candidate that maximize the selected selectivity index, which is assessed using the afore-described simulation tool [36] (i.e., considering all the possible FLs). Thus, the selected FL is permanently included in the set the constraints, and the process of selection of a new FL is repeated until reaching the desired level of selectivity. As the applied OC-DOCP minimizes a weighted sum of tripping times, indirectly maximizes the protection speed and, consequently, the desired features of speed and selectivity for the protective system are being maximized simultaneously.

This paper presents a Monte Carlo method to solve the OC-DOCP considering: (a) all the possible FLs in the transmission system; (b) the TC-STABLE; (c) the multi-objective nature of the coordination of protections. The use of statistical-based performance indexes for the OC-DOCP is a novelty of the proposed method. A key contribution of this paper is the selection of the reduced set of FLs which is necessary to solve the OC-DOCP maximizing the selectivity index. After the inclusion of the selected FLs by the Monte Carlo method, the results show that an important increase in the selectivity index is found without an important degradation of the speed index, and this is another key contribution of this paper.

This paper is organized as follows. Section 2 summarizes the formulation of the OC-DOCP considering TC-STABLE. Section 3 briefly describes the simulator used to compute performance indexes for selectivity and speed. The proposed method is presented in Section 4. Results obtained from a test system is shown in Section 5, and the conclusion is presented in Section 6.

2. OC-DOCP considering TC-STABLE

The inclusion of the constraints related to Transient Configurations due to the Sequential Tripping At Both Line-Ends (TC-STABLE) in the OC-DOCP was firstly proposed many years ago by [18], and it has been also later considered in other publications [5,12,20,24,27–29,34]. Herein, the technique described in [28] was applied, in conjunction with Monte Carlo simulation, to develop the proposed method.

A description of the technique shown in [28], using the nomenclature shown in [36], is included in Appendix A. In this section, a very summarized description of that method is shown, for the sake of readability. The mathematical problem is formulated as the search of the time dial settings ($\mathbf{D}$) that minimize the sum of operation times of

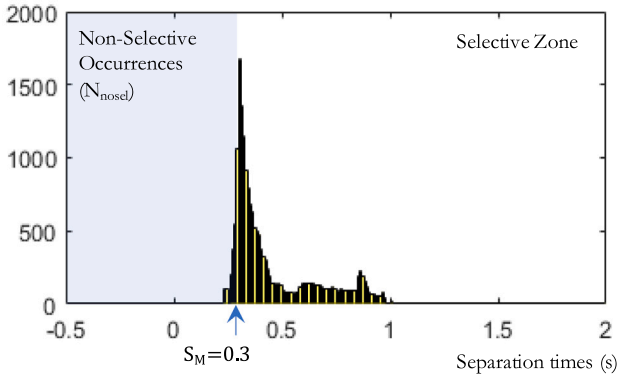


Fig. 1. Histogram of separation times obtained by the simulator [36].

primary DOCPs (T , Eq. (1)) subject to selectivity constraints ($S \geq S_M$, Eq. (2)), and upper and lower bounds of $\mathbf{D}$ (Eq. (3)):

$$\min T = \sum_{r=1}^{n_r} T_{r,c} \quad (1)$$

subject to:

$$\mathbf{S} = \mathbf{B}\mathbf{D} \geq \mathbf{S}_M \quad (2)$$

$$\mathbf{D}_{\min} \leq \mathbf{D} \leq \mathbf{D}_{\max} \quad (3)$$

where $T_{r,c}$ is the operation time of main DOCP r for the close-in fault c . $\mathbf{S}$ is the vector of computed separation times, considering all the main-backup pairs for all the fault locations included in the optimization model. Each computed separation time is the operation time of the backup DOCP minus the operation time of the main DOCP for the correspondent fault location. All the DOCP operation times are computed according to the curve equation, pick-up current, and fault current seen by the correspondent DOCP. $\mathbf{S}_M$ is a vector whose elements are the safety margin S_M .

This formulation is identical than the basic one that does not consider TC-STABLE. In order to consider TC-STABLE, the technique described in [28] includes the effect of the line-end which firstly trips in the definition of the relationships among the time dial adjustments (i.e., in the matrix $\mathbf{B}$). The details about the structure of the coordination matrix $\mathbf{B}$ considering TC-STABLE can be found in [36].

Actually, the sequence of DOCP trips is dependent on the DOCP settings. In order to obtain coherency, the solution of the OC-DOCP must have the same sequence of DOCP trips than the assumed one to build the set of constraints. This problem is solved by an iterative procedure, until reaching convergence [28]. The initial guess about the sequence of DOCP trips is obtained from an OC-DOCP without considering TC-STABLE (as the one originally formulated in [3]).

The information about the currents seen by each DOCP (in multiples of the DOCP pickups), for each fault location and considering the TC-STABLE, is in the matrix $\mathbf{B}$. Herein, it is assumed that the sensitivity problem has been previously maximized (i.e., the DOCP pickups have been previously selected to avoid wrong trips due to normal load conditions), as it has been considered by a subset of the literature about OC-DOCP. On the other hand, the information about the curve type also is in the matrix $\mathbf{B}$ since it defines the relationship between the current and the operation time in each DOCP, for each analyzed fault.

Thus, the unknown variables of the OC-DOCP are the dial time settings ($\mathbf{D}$) and the basic data needed to solve this problem are:

1. The vector of fault locations to be considered.
2. Fault currents seen by each DOCP of each pair main-backup, for each FL.
3. Pickup current and curve type for each DOCP.

4. Safety margin (S_M) for the selectivity, defined by the protection engineer. This margin is also called “coordination time interval”, but this term is not applied here in order to avoid confusions (because coordination is much more than selectivity).

3. Simulator to assess statistical performance indexes for selectivity and speed

The coordination of DOCP has been usually performed considering very few Fault Locations (FLs). In order to assess the performance of any coordination of DOCP (based on optimization tools or not), a simulator was previously developed [36]. This simulator assesses performance indexes of speed, selectivity, and sensitivity using a high number of FLs and considering the TC-STABLE. This simulator is a key part of the method herein proposed. The simulator needs the DOCP adjustments as input data, which are the overcurrent pickups, the time-overcurrent curve types and the time dial settings ($\mathbf{D}$). This simulator computes the fault currents seen by each DOCP for a total of n_{fls} uniformly distributed FLs in each transmission line. The FLs are defined in per unit of line lengths (e.g., 0.5 means fault at the midpoint of each analyzed line). For each FL, the operation times of the main DOCPs at each line-end of the faulted line are computed, considering the TC-STABLE, as well as the operation times of the backup DOCPs of each pair main-backup. In some cases, the directional and/or the overcurrent elements of the backup DOCP cannot operate and, consequently, a separation time between main and backup DOCPs cannot be computed; in the remaining cases, the vector $\mathbf{S}^+$ of computed separation times is given by Eq. (4):

$$\mathbf{S}^+ = \mathbf{B}^+ \mathbf{D} \quad (4)$$

Vectors $\mathbf{S}^+$ and $\mathbf{B}^+$ are similar to the previously defined $\mathbf{S}$ and $\mathbf{B}$, but: (a) they are not part of an optimization since the simulator only assesses the results for a given solution $\mathbf{D}$; (b) their dimensions are notoriously bigger since the number of considered FL is high enough in order to guarantee an assessment for practically all the possible FLs. The number of considered FLs for the simulator is n_{fls} , whereas the number of FLs to be considered in the OC-DOCP constraints is n_{sfl} .

The set of cases whose separation time can be computed is N_F . N_{nosel} is the subset of $\mathbf{S}^+$ whose computed separation time is lower than the safety margin S_M , and Eq. (5) defines the selectivity index:

$$\text{selectivity index} = 1 - \frac{N_{nosel}}{N_F} \quad (5)$$

As an example, a histogram of the computed separation times $\mathbf{S}^+$, using the published results of an OC-DOCP method [37], is shown in Fig. 1 (additional details of this example are described in Section 5). The histogram shows the number of occurrences of computed separation times for n_{fls} simulated FLs, in ranges of 0.01 s (also called bins), from -0.5 to 2 s. The sum of all the occurrences is N_F . The number of occurrences in the shadowed area N_{nosel} is 2392, N_F is 19548, and the selectivity index is 87.8%.

$T_{r,i}^+$ is the simulated operation time of main DOCP r for the fault location i , the correspondent speed is defined as $1/T_{r,i}^+$, and the speed index simply is the average speed:

$$\text{speed index} = \frac{1}{n_r n_{fls}} \sum_{r=1}^{n_r} \sum_{i=1}^{n_{fls}} \frac{1}{T_{r,i}^+} \quad (6)$$

Although a high number of FLs is included in this simulator, the behavior of the DOCPs for each analyzed FL can be independently assessed and, consequently, the simulations can be carried out using parallel computing in order to reduce the total simulation time. This is a key point, because the high number of FLs is not included in the optimization stage of the proposed method.

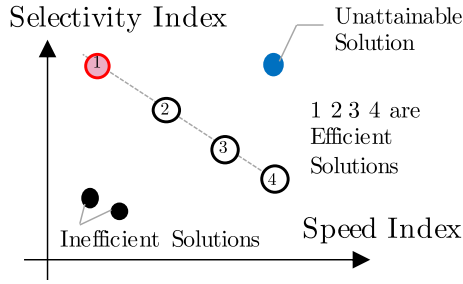


Fig. 2. Trade-offs among different OC-DOCP solutions.

4. Proposed method

4.1. Solution strategy

The coordination of protections intrinsically is a multi-objective problem; only sensitivity, selectivity, and speed have been the objectives to be maximized in the OC-DOCP, for the sake of simplicity [39]. Herein, the sensitivity is assumed as a previously solved problem; in this context, an increase of selectivity implies a decrease of speed for the solutions of the OC-DOCP, as it is illustrated in Fig. 2.

Although different speed and selectivity indexes could be chosen, the statistical indexes described in Eqs. (5) and (6) are applied here. Thus, the conceptual formulation of the multi-objective optimization problem, using these selectivity and speed performance indexes, is:

$$\max \text{ selectivity index} \quad (7)$$

$$\max \text{ speed index} \quad (8)$$

subject to:

$$S = BD \geq S_M$$

$$D_{\min} \leq D \leq D_{\max}$$

This problem can be reformulated as a single-objective optimization problem using the ϵ -constraint solution strategy [40]:

$$\max \text{ speed index}$$

subject to:

$$\text{selectivity index} \geq \epsilon_{sel} \quad (9)$$

$$S = BD \geq S_M$$

$$D_{\min} \leq D \leq D_{\max}$$

A mathematically rigorous solution for this problem needs to face two difficulties: (a) the speed index is the average of the DOCP speeds, and each speed is related to the inverse of the tripping time, which introduces an important nonlinearity in this objective function; (b) the selectivity index is based on counting the number of cases without selectivity for a given solution of D , which is the vector of decision variables.

The proposed method is an iterative algorithm that includes only a limited number (n_{sfl}) of FLs in the set of OC-DOCP constraints. These FLs are selected using Monte Carlo simulation in order to find the best FL to be included in each step, from a set of n_c randomly generated candidates. Two FLs (near-end, far-end) are included in the set of OC-DOCP constraints from the beginning.

The OC-DOCP described in Section 2 is solved, including each candidate at once, and each optimal result D is assessed using the simulation tool described in Section 3. Subsequently, the best FL is selected as the candidate with maximum selectivity index. This selected FL is permanently included in the set the constraints, and the process of selection of a new FL is repeated until reaching the desired level of selectivity (ϵ_{sel}). The highest possible level of selectivity is 100%. As

the applied OC-DOCP minimizes a weighted sum of tripping times, indirectly maximizes the protection speed and, consequently, the desired features of speed and selectivity for the protective system are being considered simultaneously.

The solution strategy comprises four steps:

1. Determine an initial solution for the OC-DOCP with only two FLs (Section 2). Assess the selectivity index by running the simulator (Section 3).
2. Generate a candidate FL (insert a random FL between $\{0,1\}$). Determine a new solution for the OC-DOCP including the constraints due to this candidate FL (Section 2). Assess the selectivity index by running the simulator (Section 3).
3. Repeat the step 2 for the number of candidates n_c , and select the FL with highest selectivity index to be permanently included in the set of constraints.
4. If an additional improvement of the selectivity index is necessary, repeat steps 2 and 3 to select a new FL to be permanently included in the set of constraints.

Any other traditional OC-DOCP formulation can be applied in conjunction the described simulator, since the traditional OC-DOCP maximizes the protection speed whereas the combination of the Monte Carlo method with the described simulator maximizes the selectivity index. On the other hand, if other simulators are developed, keeping the idea of assessing all the FLs and considering the TC-STABLE, any of them could be applied in combination with the described Monte Carlo method to maximize the selectivity index from the OC-DOCP solutions. That is, the developed method is general and it can be applied with different tools

4.2. Details of the developed method

A detailed description of the developed algorithm is shown in Algorithm 1. The first step is an OC-DOCP (Section 2) with only two FLs (near-end and far-end). The counter k is the number of FLs that are included in the set of OC-DOCP constraints. The first iteration is for $k=3$, but the initial results (for $k=2$) are also computed and stored. In each iteration for k , n_c FL candidates are randomly generated (j counts the number of generated candidates) in order to assess each FL candidate using the OC-DOCP and the simulator, as described in Sections 2 and 3, respectively. Once the n_c FL candidates have been assessed, the FL candidate with the best selectivity index is selected to remain in the set of OC-DOCP constraints, the correspondent results are stored, and a new iteration begins (increasing the counter k). The iterative process finishes when the computed selectivity index is greater than the desired level of selectivity ϵ_{sel} . The results for each iteration k (selected FL, optimal value of D , selectivity and speed indexes) are stored, using a subscript equal to the number of FLs included in the set of OC-DOCP constraints. The selected FL for each iteration k is stored as a new element of the vector H for this purpose. The number of selected FLs is n_{sfl} , and it is also the final the number of elements of H (although the iterations begin at $k=3$, because the near-end and far-end are also “selected FL” using this nomenclature).

That is, the Monte Carlo simulation has two loops. The external loop (steps 7–23) stops when the specified ϵ_{sel} is reached. The internal loop (steps 9–15) runs n_c times, to generate and assess n_c random FL in order to select the one with best selectivity index. If this selectivity index is not better than the obtained one in the previous iteration, the internal loop is executed again without increasing the external loop counter.

An adequate selection of the number of candidates n_c is very important. In the performed tests, the use of $n_c=100$ was an excellent option to achieve solutions with high level of selectivity and reasonable computing times. The use of very low values of n_c implied serious difficulties to reach high levels of selectivity, and the use of very

high values of n_c implied excessive computing times without important improvements in the final results.

Algorithm

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1: Inputs:
   Power system data, protection system data
   Number of candidate FLs  $n_c$ ,
   Desired selectivity index  $\epsilon_{sel}$ 
2: Set  $k \leftarrow 2$  Two FLs (near-end far-end)
3: Run OC-DOCP (Section 2)
4: Set  $\mathbf{D}_2$  as an initial coordination solution
5: Run the Simulator for  $\mathbf{D}_2$  (Section 3)
6: Store intermediate result  $selectivity_2$ 
7: while  $selectivity_k \leq \epsilon_{sel}$  do
8:    $k=k+1$  [ $k$ -th selected FL]
9:   for  $j=1$  to  $n_c$  do [ $n_c$  candidates]
10:     $H_{k,j} \leftarrow$  insert a random FL between 0 and 1 pu
11:    Run OC-DOCP (Section 2)
12:    Store result  $\mathbf{D}_{k,j}$  as candidate
13:    Run the Simulator for  $\mathbf{D}_{k,j}$  (Section 3)
14:    Store results  $\mathbf{S}_{k,j}^+$ ,  $speed_{k,j}$ ,  $H_{k,j}$ , and  $selectivity_{k,j}$ 
15:   end for
16:   Store intermediate result  $selectivity_k \leftarrow \max(selectivity_{k,j})$ 
17:   Store intermediate result  $speed_k \leftarrow speed_{k,j}$  [case of  $\max(selectivity_{k,j})$ ]
18:   Store intermediate result  $\mathbf{D}_k \leftarrow \mathbf{D}_{k,j}$  [case of  $\max(selectivity_{k,j})$ ]
19:   Store intermediate result  $\mathbf{S}_k^+ \leftarrow \mathbf{S}_{k,j}^+$  [case of  $\max(selectivity_{k,j})$ ]
20:   Store intermediate result  $H_k \leftarrow H_{k,j}$  [case of  $\max(selectivity_{k,j})$ ]
21:   if  $selectivity_k \leq selectivity_{k-1}$  then  $k = k - 1$  go to 9
22:   end if
23: end while [Loop stops when  $selectivity_k$  exceeds  $\epsilon_{sel}$ ]
24: Store final result  $selectivity \leftarrow selectivity_k$ 
25: Store final result  $speed \leftarrow speed_k$ 
26: Store final result  $\mathbf{D} \leftarrow \mathbf{D}_k$ 
27: Outputs:
   Vector of  $n_{sfl}$  selected FLs,  $\mathbf{H} = [H_1, \dots, H_{n_{sfl}}]$ 
   Coordination solutions  $\mathbf{D}_3, \dots, \mathbf{D}_{n_{sfl}}$ 
   ( $selectivity_3, speed_3, \dots, (selectivity_{n_{sfl}}, speed_{n_{sfl}})$ )

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Algorithm 1: Description of the method

5. Results

A well-known system, originally shown in [38], is taken as an example (Fig. 3). The following conditions are included here as well as in two previous publications [28,37] taken as examples to compare results: (a) standardized IEC curves are considered; (b) all the DOCP are set with the same curve type; (c) the safety margin (S_M) is 0.3 s; (d) ground fault functions are not considered; (e) only three-phase solid faults are considered; (f) the pickup currents of the DOCP are the original values shown in [37], which were selected to maximize sensitivity considering the normal load currents. Results with Standard Inverse (SI) curves are compared with those from both previous publications [28,37], and results with Very Inverse (VI) and Extremely Inverse (EI) curves are only compared with those from [28] because [37] does not show results for VI nor EI curves. These results are shown in Section 5.1.

On the other hand, results with SI curves from another publication [35] are also taken for comparison purposes. The main differences with the previous example are: (a) the pickup currents shown in [35] are much smaller than the original values shown in [37]; (b) none of the three objective functions shown in [35] is the speed maximization; (c) the safety margin (S_M) is 0.2 s. Due to these important differences, these results are shown in Section 5.2 for the sake of clearness.

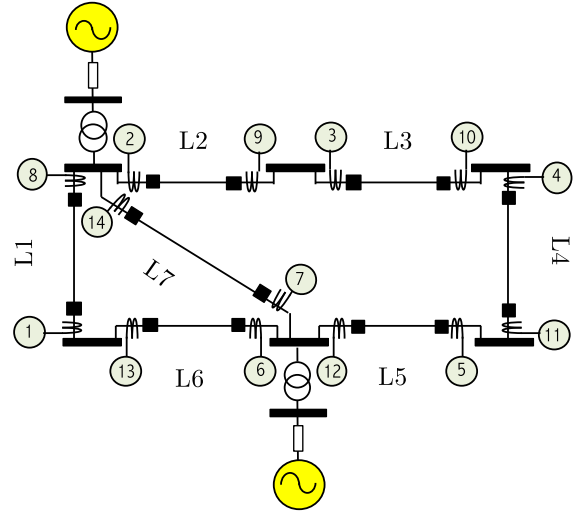


Fig. 3. Test system (data taken from [38])

5.1. Comparison with previously reported OC-DOCP solutions that maximize speed keeping maximum sensitivity

The simulator is set to $n_{fls}=1000$ (uniformly distributed faults locations) to assess the previously published OC-DOCP solutions as well as to be part of the proposed method. As the number of DOCP pairs of the system is 20, a total of 20000 combinations are assessed by the simulator for each vector of time dial adjustments ($\mathbf{D}$). In the proposed method, the Monte Carlo simulation was set to analyze 100 FL candidates (n_c) in each iteration k , and the desired selectivity index (ϵ_{sel}) was set to 99.9%. In order to enhance the understanding of the proposed method, a numerical example of arrays $\mathbf{S}$, $\mathbf{B}$, $\mathbf{D}$ and $\mathbf{S}_M$ is shown in Appendix B.

5.1.1. Convergence of the proposed method

With the developed method, the selectivity index reaches 99.94%, 99.95% and 99.96%, after selecting 18, 21 and 23 FL, for SI, VI and EI curves, respectively. Fig. 4 shows the reached selectivity indexes as a function of the number of selected FLs for each curve type. With ϵ_{sel} equal to 99%, the number of selected FLs should be in the order of 13, whereas with ϵ_{sel} equal to 95%, the number of selected FLs should be in the order of 6.

Fig. 5 shows the simultaneous evolution of the speed and selectivity indexes for each curve type. In these graphs, the first iteration is for 3 FLs, and the number of selected FLs is increased until the obtaining of $\epsilon_{sel}=99.9\%$. Due to the multi-objective nature of this problem, an increase of the selectivity index implies a decrease of the speed index. However, the total increase of the selectivity indexes is in the order of 7%, 12% and 15%, whereas the total decrease of the speed indexes is in the order of 2%, 4% and 5%, for SI, VI and EI curves, respectively. That is, the increment of selectivity is considerably higher than the decrease of speed, using these statistical indexes that consider all the FLs, and this fact points toward preferring the increment of selectivity. This behavior probably occurs because the initial solution, found with FLs at near-end and far-end, is already relatively near of the final solution for $\mathbf{D}$ despite the lack of selectivity for relatively high number of FLs.

Fig. 5 is also useful to illustrate the ϵ -constraint solution from a different perspective. Assume, for example, the use of SI curve and a ϵ_{sel} target equal to 99%; Fig. 5(a) shows that different solutions with selectivity index greater than 99% are feasible, but there is one that maximizes the speed (the detailed results for this solution are: 12 selected FLs, selectivity index=99.16%, speed index=1.0471 s⁻¹).

Table 1

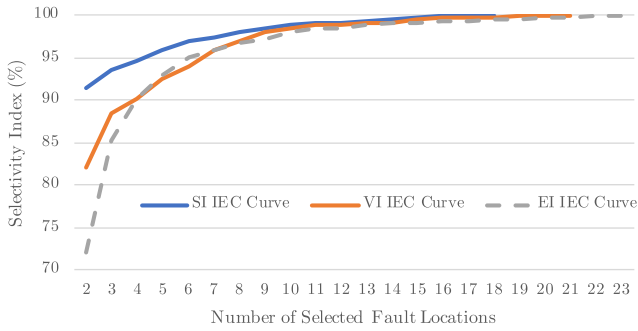
Selectivity and speed indexes for two previously reported OC-DOCP solutions and the proposed method (SI curve).

	Curve type	Existing optimal DOCP methods		Proposed method
		[37]	[28]	
Consider TC-STABLE?		no	yes	yes
Number of FLs ($n_{s,fl}$)	SI	1 ^a	3 ^b	18 ^c
Selectivity index (%)	SI	87.8	89.5	99.94
Speed index (s^{-1})	SI	1.154	1.089	1.045

^a Fault Location (FL): Near-end.^b FL: Near-end, far-end, and line mid-point.^c FL: Near-end, far-end, and the selected FLs by the proposed method.**Table 2**

Selectivity and speed indexes for a previously reported OC-DOCP solution and the proposed method (VI and EI curves).

	Curve type	Existing optimal DOCP methods		Proposed method
		[28]		
Consider TC-STABLE?		yes		yes
Number of FLs ($n_{s,fl}$)	VI	3 ^a		21 ^b
	EI	3 ^a		23 ^b
Selectivity index (%)	VI	84.70		99.95
	EI	80.10		99.96
Speed index (s^{-1})	VI	2.314		2.219
	EI	4.256		3.981

^a FL: Near-end, far-end, and line mid-point.^b FL: Near-end, far-end, and the selected FLs by the proposed method.**Fig. 4.** Convergence path of the selectivity index (SI, VI, and EI curves)

5.1.2. Comparison with the previously reported OC-DOCP solutions

In case of SI curves, **Table 1** shows that the selectivity indexes for the optimal solutions from two previously reported methods are 87.8% for [37] and 89.50% for [28]. That is, an improvement in the order of 12% for the selectivity index is obtained by the proposed method, with a worsening for the speed index in the order of 7%.

In case of VI and EI curves, **Table 2** shows that the selectivity indexes for the optimal solutions previously reported in [28] are 84.70% and 80.10%, respectively. That is, an improvement in the order of 20% for the selectivity index is obtained by the proposed method, with a worsening for the speed index in the order of 5%.

5.1.3. Effect of rounding the results

Sections 5.1.1 and 5.1.2 describe results from the proposed method using the full mathematical resolution for the adjustment of time dial settings (**D**). Such results are useful to explore the method mathematical performance as well as to show the method's capacity to reach values very close to 100% of selectivity.

Although the numerical relays could theoretically have the availability of very high resolution for these adjustments, the use of more than three decimals for the time dial settings of IEC curves is not common at all in practice (the use of more resolution is not compatible with the different possible sources of errors, such as current transformers' accuracy, system impedances, and other factors). Due to this fact, it is important to analyze the effect of rounding the adjustments' values.

Table 3 shows the results of selectivity indexes after the decimal rounding of the adjustments computed to obtain $\epsilon_{sel}=99.9\%$. These results imply that the target of $\epsilon_{sel}=99.9\%$ tends to be excessive, considering the shown options for decimal rounding of **D**. For example, if the adjustments will be rounded to 3 decimals, then: (a) an option is the use of $\epsilon_{sel}=99.9\%$ and the results shown in **Table 3**; (b) another option is the use of a lower target (e.g., $\epsilon_{sel}=99\%$) and the correspondent numbers of selected FLs to reach that target would be significantly reduced. The first option is adequate if the computing time is not a concern; otherwise, the second option seems to be more attractive. There is also a third option which is the search of the combinatorial optimum of discrete adjustments, considering the decimal rounding; perhaps this option is mathematically interesting, but surely is not justifiable in practice.

5.1.4. Comparison with results of using uniformly distributed faults

The number of selected FLs required to reach $\epsilon_{sel} \geq 99.9\%$ for SI, VI and EI curves are 18, 21 and 23, respectively (these results were shown in Sections 5.1.1 and 5.1.2). If the same numbers of FL were uniformly distributed, the results of selectivity and speed indexes would be considerably different, as shown in **Table 4**. The selectivity indexes reached with the proposed method are notoriously nearer to 100% than those reached using uniformly distributed faults, and the worsening of the speed index is not significant.

5.1.5. Computational details of the implemented method

The developed method was implemented using Matlab 2023b interpreter in an Apple MacBook Pro (MacOS Monterrey) Quad-Core Intel Core i7 2.6 GHz with 16 MB RAM. The linear programming problem was solved using linprog, which is an interior point algorithm available in Matlab. The CPU times required to obtain the solutions with SI curves were: (a) with one core: 984 s (minimum), 1031 s (average), 1101 s (maximum); (b) with four cores: 256 s (minimum), 267 s (average), 285 s (maximum).

5.2. Comparison with other previously reported OC-DOCP solutions

Another publication [35] was taken for comparison purposes in this section, but such a publication does not consider the original pickup currents shown in [37], and does not consider the protection system speed as the objective function. **Table 5** shows the results from

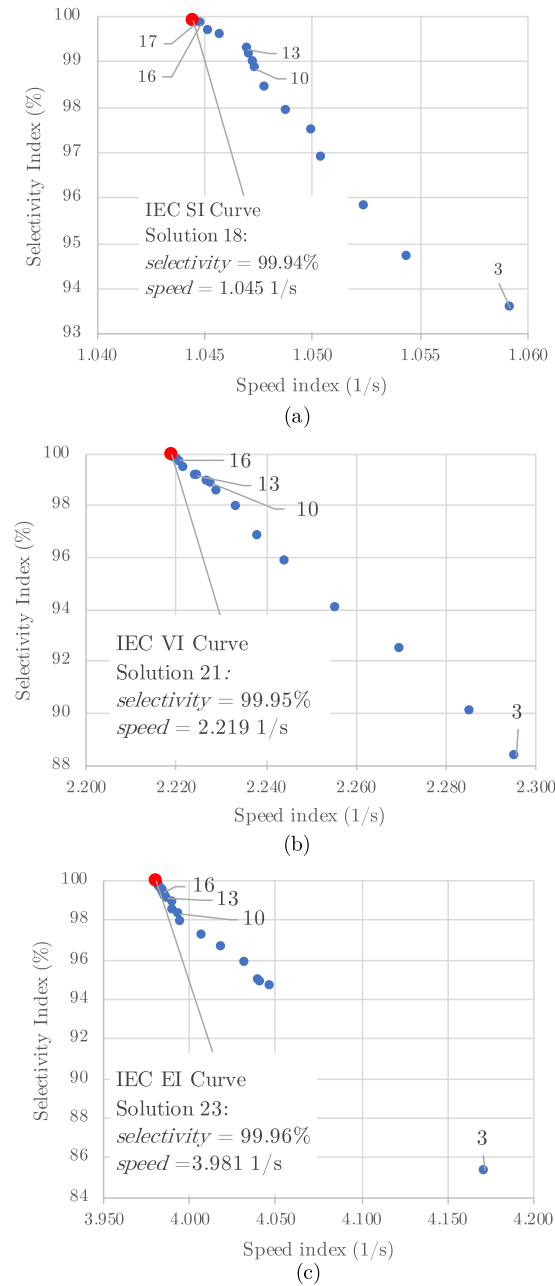


Fig. 5. Simultaneous evolution of selectivity and speed indexes.

the simulator with the pickup currents and time dial settings shown in [35]; the curve type is IEC SI, and the safety margin (S_M) is 0.2 s. The three cases of time dial settings shown in [35] correspond to different objective functions. On the other hand, the optimal results were obtained with the proposed method and are also shown in Table 5.

The results from [35] are far from the optimal, since correspondent speed indexes are much smaller than the optimal value. That is, all the results from [35] imply very slow protections. This fact is a logical consequence of no having the speed as an essential objective in the problem formulation shown in [35]. However, the most interesting result is that the case 3 from [35] reaches 100% of selectivity, despite the fact of no considering TC-STABLE nor all the FLs in the problem formulation. This is a surprising result since other OC-DOCP formulations, apparently better supported, had seen assessed [36] and none of them reached selectivity indexes close to 100%. This surprising result seems to be a very strange coincidence. The main hypothesis about this surprising

result is that the objective function for case 3 of [35] tends to separate very much the resulting operation times and, coincidentally, this fact implies 100% of selectivity in this example. This surprising result could be further analyzed in the future since nice findings could sometimes arise from this type of unexpected results.

6. Conclusion

The use of statistical-based performance indexes for the optimal coordination of directional overcurrent protections is a novel technique herein proposed. The developed method can guarantee selectivity for all the fault locations, considering also the transient configurations due to the sequential tripping at both line-ends. The use of Monte Carlo simulation in the proposed method is the key point to avoid the inclusion of a huge number of constraints related to all the possible fault locations, by selecting only a few of relevant fault locations for the

Table 3Effect of decimal rounding, of adjustments computed to obtain $\epsilon_{sel}=99.9\%$, on selectivity indexes).

D	SI			VI			EI		
	Full resolution	Three decimals	Two decimals	Full resolution	Three decimals	Two decimals	Full resolution	Three decimals	Two decimals
D_1	0.37178	0.372	0.37	0.37201	0.372	0.37	0.53131	0.531	0.53
D_2	0.35982	0.360	0.36	0.40267	0.403	0.40	0.49157	0.492	0.49
D_3	0.37214	0.372	0.37	0.55889	0.559	0.56	1.16805	1.168	1.17
D_4	0.36080	0.361	0.36	0.62524	0.625	0.63	1.96248	1.962	1.96
D_5	0.22454	0.225	0.22	0.16087	0.161	0.16	0.13683	0.137	0.14
D_6	0.44725	0.447	0.45	0.68173	0.682	0.68	1.86273	1.863	1.86
D_7	0.50120	0.501	0.50	0.76524	0.765	0.77	1.35771	1.358	1.36
D_8	0.44052	0.441	0.44	0.67792	0.678	0.68	1.86295	1.863	1.86
D_9	0.22183	0.222	0.22	0.15932	0.159	0.16	0.13670	0.137	0.14
D_{10}	0.35797	0.358	0.36	0.62313	0.623	0.62	1.96273	1.963	1.96
D_{11}	0.37220	0.372	0.37	0.55781	0.558	0.56	1.17007	1.170	1.17
D_{12}	0.32500	0.325	0.33	0.32333	0.323	0.32	0.33820	0.338	0.34
D_{13}	0.36572	0.366	0.37	0.36744	0.367	0.37	0.53183	0.532	0.53
D_{14}	0.49324	0.493	0.49	0.75618	0.756	0.76	1.35393	1.354	1.35
Selectivity (%)	99.94	99.07	94.85	99.95	99.60	98.43	99.96	99.58	97.36

Table 4

Comparison between the results from the proposed method and the results obtained with the same numbers of uniformly distributed faults.

Curve type	Selectivity index (%)		Speed index (s ⁻¹)	
	Uniformly distributed FLs	Proposed method	Uniformly distributed FLs	Proposed method
SI	97.53	99.94	1.052	1.045
VI	98.75	99.95	2.226	2.219
EI	98.76	99.96	3.994	3.981

Table 5

Selectivity and speed indexes with SI curves for the three cases shown in [35] and for the optimal solution.

DOCP number	Pick-Up currents ^a (kA)	Time Dial settings (D)			
		Case 1	[35] Case 2	Case 3	Proposed method
1	0.125	0.450	0.600	0.666	0.242
2	0.200	0.700	0.650	0.626	0.335
3	0.150	0.600	0.550	0.554	0.270
4	0.200	0.450	0.400	0.388	0.169
5	0.137	0.250	0.350	0.365	0.132
6	0.137	0.650	0.750	0.983	0.333
7	0.162	0.550	0.600	0.650	0.293
8	0.137	0.500	0.800	0.893	0.359
9	0.135	0.200	0.350	0.398	0.142
10	0.137	0.350	0.500	0.499	0.233
11	0.162	0.400	0.550	0.586	0.310
12	0.137	0.550	0.750	1.000	0.432
13	0.150	0.300	0.600	0.558	0.248
14	0.200	0.450	0.500	0.638	0.300
Selectivity index (%)		94.1	96.8	100.0	100.0
Speed index (s ⁻¹)		0.947	0.728	0.672	1.564
Avg. Op. Time (ms)		1105.9	1398.0	1537.1	660.0

^a These pickup currents do not consider the maximum possible sensitivity due to the load currents in the original system [37], and they are included here only to compare with results shown in [35].

set of constraints. The use of multi-objective formulation, considering selectivity and speed indexes as simultaneous objectives to be maximized, offers a proper perspective of this problem which is featured by the trade-off between both objectives.

The proposed method was applied to a well-known sample system, whose solutions from two previous articles only reach selectivity indexes between 80% and 90%, and the solutions herein shown assure a selectivity index greater than 99.9%. As the solutions of the first iteration of the proposed algorithm (which reach selectivity indexes greater than 85% using only three fault locations) are very close to the final solutions from the perspective of the speed index, the obtained increments in the selectivity index seem to be completely justifiable.

CRediT authorship contribution statement

Paulo M. De Oliveira-De Jesus: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Elmer Sorrentino:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Summary of OC-DOCP considering TC-STABLE [28, 36]

Fig. A.6 shows a generic power system taken as an example for this explanation. Different possible faults locations H_i are shown. n_{fl} is the number of considered fault locations. For any fault location H_i , the DOCP at both line-ends should trip, q is the DOCP that firstly trips, and the DOCP at the other line-end is not herein described for the sake of brevity. DOCP p is one of the backup protections for DOCP q , and the fault currents seen by DOCP q and p for H_i are $I_{q,i}$ and $I_{p,i}$, respectively. The DOCP operation times depend on the curve equation, pick-up current, and fault current seen by the correspondent DOCP. For instance, in case of IEC standardized curves, their equations have the form described in Eq. (A.1), for a generic DOCP r and for a fault i .

$$T_{r,i} = \frac{\alpha_{r1} D_r}{[\frac{I_{r,i}}{P_r}]^{\alpha_{r2}} - 1} \quad (\text{A.1})$$

D_r is the time dial setting of DOCP r ; $T_{r,i}$ is the operation time of DOCP r for fault i ; α_{r1} and α_{r2} are constants defined by the standardized IEC curve type; P_r is the pickup current of DOCP r ; $I_{r,i}$ is the fault current seen by DOCP r for the fault location H_i . Eq. (A.1) is only an example of the possible forms of the equations that describe the DOCP operation times.

For a given power system, there are n_{pr} pairs of backup-main DOCP. For each DOCP pair u and fault i , a separation time $S_{u,i}$ can be calculated ($S_{u,i}$ is the time difference between operation times of backup and primary DOCP). Each separation time must be greater

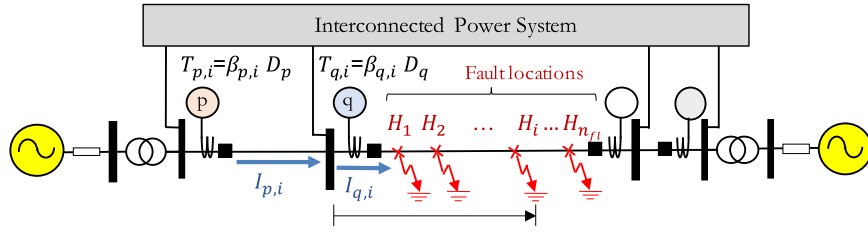


Fig. A.6. Sketch to illustrate the explanation about OC-DOCP fundamentals.

than a given safety margin S_M , specified by the protection engineer (selectivity constraints, described by Eq. (A.2)).

$$S_{u,i} = T_{p,i} - T_{q,i} = \beta_{p,i} D_p - \beta_{q,i} D_q \geq S_M \quad (\text{A.2})$$

$\beta_{p,i}$ and $\beta_{q,i}$ are linear coefficients, which depend on curve types, pickup currents and fault currents seen by each DOCP. Eqs. (A.3) and (A.4) show examples of these coefficients, using the particular case of IEC standardized curve types (described by Eq. (A.1)).

$$\beta_{p,i} = \frac{\alpha_{p1}}{[\frac{I_{p,i}}{P_p}]^{\alpha_{p2}} - 1} \quad (\text{A.3})$$

$$\beta_{q,i} = \frac{\alpha_{q1}}{[\frac{I_{q,i}}{P_q}]^{\alpha_{q2}} - 1} \quad (\text{A.4})$$

The separation time $S_{u,i}$ for each DOCP pair u and fault i , using Eq. (A.2), defines the term of the coordination matrix $\mathbf{B}$, shown in Eq. (2). That is, the elements of $\mathbf{B}$ should be determined considering the fault currents in main DOCP and backup DOCP, and each row of matrix $\mathbf{B}$ only has two elements. If the sensitivity of a backup DOCP is unable to detect the fault current ($\beta_{p,i} \leq 0$), or the fault current $I_{p,i}$ is not in the trip direction, then the corresponding row for this DOCP pair must not be included in the coordination matrix $\mathbf{B}$.

On the other hand, the described structure for $\mathbf{B}$ does not consider the transient configurations due to the sequential tripping at both line-ends. The details of a simple and convenient way to include this effect can be found in [28]. A simple difference regarding the described structure for $\mathbf{B}$ is the fact of having three elements in the row of the matrix $\mathbf{B}$ for the DOCP of the line-end that trips later, because the selectivity constraints depend on the main DOCP, the backup DOCP and the DOCP of the line-end that firstly trips [28,36]. A description with nomenclature similar to the one of this Appendix can be found in [36] (where the additional linear elements of the matrix $\mathbf{B}$ are denominated γ); these details are not herein included for the sake of conciseness. For a given fault, a backup DOCP can be insensitive for the first network topology (before the operation of the line-end that firstly trips), and not for the second network topology (after the operation of the line-end that firstly trips); the corresponding row for that DOCP pair must be included in the coordination matrix $\mathbf{B}$ with the proper equations for that case. However, if a backup DOCP is insensitive for a given fault during the second network topology, and the corresponding row for the DOCP pair must not be included in the coordination matrix $\mathbf{B}$, as described in the previous paragraph.

Appendix B. Numerical example of matrix $\mathbf{B}$ and vectors, $\mathbf{S}$ and $\mathbf{S}_M$

The numerical values of the terms of the coordination matrix $\mathbf{B}$ and the separation time vector $\mathbf{S}$ depend on curve type, pickup currents, fault currents, and main-backup pairs. This numerical example corresponds to a model with TC-STABLE (details in Appendix A) considering SI curves, with pickup currents that maximize sensitivity according to the maximum load currents (given in [37]) of the test case [38]. For any given fault location FL, the numerical values of vectors $\mathbf{B}$ and $\mathbf{S}$ change. For instance, for a FL located very near one line-end, this FL is the case of far-end for the other line-end. This example has 7 transmission

lines; thus, in case of the FL very close to the DOCP 1, 2, 3, 4, 5, 6, and 14, the corresponding far-ends are the DOCP 8, 9, 10, 11, 12, 13, and 7, respectively. The rows of the matrix $\mathbf{B}$, corresponding to the 20 main-backup pairs, are in the following order: 2-1, 2-7, 3-2, 4-3, 5-4, 6-5, 6-14, 1-6, 14-9, 14-1, 9-10, 10-11, 11-12, 12-13, 12-14, 13-8, 7-5, 7-13, 8-9, and 8-7. Thus, $\mathbf{B}$ is:

$$\mathbf{B} = \begin{bmatrix} 3.592 & -2.875 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -2.875 & 0 & 0 & 0 & 0 & 0 & 2.673 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3.612 & -2.587 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3.063 & -2.321 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2.813 & -2.813 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5.557 & -2.037 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -2.037 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2.673 \\ -2.178 & 0 & 0 & 0 & 0 & 0 & 2.514 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 3.592 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 5.557 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1.891 \\ 0 & 0.000 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1.891 \\ 0 & 0 & 0.160 & 0 & 0 & 0 & 0 & 0 & 0 & 2.213 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.187 & 0 & 0 & 0 & 0 & 0 & -2.098 & 2.691 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.816 & 0 & 0 & 0 & 0 & 0 & -2.515 & 3.836 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.518 & 0 & 0 & 0 & 0 & 0 & 0 & -3.430 & 3.826 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.119 & 0 & 0 & 0 & 0 & 0 & 0 & -2.005 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4.114 & 0 & -1.901 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1.345 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1.901 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2.921 & 1.345 & 0 \\ 1.799 & 0 & 0 & 0 & 0 & 0 & 0 & -2.138 & 5.443 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1.799 & 0 & 0 & 0 & 0 & 0 & 0 & 2.646 & -2.138 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Notice that some rows of $\mathbf{B}$ have only two elements, corresponding to the β values of the line-end which trips first, whereas other rows have three elements, corresponding to the linear coefficients γ of the second line-end that trips, as described in Appendix A.

After reaching the optimal solution, the optimal time dial settings are: $\mathbf{D} = [0.3718, 0.3598, 0.3721, 0.3608, 0.2245, 0.4473, 0.5012, 0.4405, 0.2218, 0.3580, 0.3722, 0.3250, 0.3657, 0.4932]$.

For this specific example, with the optimal $\mathbf{D}$ and the particular case of the described FL, the vector of separation times is: $\mathbf{S} = [0.3011, 0.3056, 0.3370, 0.3022, 0.3833, 0.3366, 0.4075, 0.3148, 0.3000, 0.4027, 0.3012, 0.3097, 0.3782, 0.4678, 0.3839, 0.3278, 0.6341, 0.7786, 0.9345, 1.0533]$ s. Vectors $\mathbf{S}$ and $\mathbf{S}_M$ have simply 20 elements in this extremely simple numerical example, because they correspond to the use of only one FL for each one of the 20 DOCP pairs (20 selectivity constraints). However, if the number of FLs considered for the constraints is higher, the correspondent dimension of $\mathbf{B}$, $\mathbf{S}$ and $\mathbf{S}_M$ will be consequently higher.

The vector $\mathbf{S}_M$ has 20 elements, and each one is the specified safety margin ($S_M = 0.3$ s, for this example). Notice that the shown elements of $\mathbf{S}$ are not less than S_M , since the value of $\mathbf{D}$ for this numerical example corresponds to the optimal solution.

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